Randy Xu 許淳荏

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Summary

AI engineer and full-stack developer. 2.5 years on Mail2000, an enterprise email platform with 10M+ licensed users; since 2024, building LLM agent applications on the Claude Agent SDK that stay verifiable and cost-aware: code-based guardrails, a per-turn cost ledger and a Python / React front end.

Skills

- **Languages**: Python, TypeScript, JavaScript, C, C#, Perl, SQL
- **AI / LLM**: Claude Agent SDK, MCP, Multi-agent orchestration, Structured output, Guardrails, Prompt caching
- **Web & Backend**: FastAPI, React, REST API design, three.js / WebGL, Jekyll
- **Databases & Infrastructure**: PostgreSQL, MongoDB, Docker, GitHub Actions, Linux, keepalived (VRRP), Git

Work Experience

AI Agent Developer Apr 2024 – Present
Independent *Remote*

hyenovel: LLM critique and discussion app for literary fiction

- **Architecture**: Orchestration layer on the Claude Agent SDK that keeps the LLM between deterministic code: the orchestrator sequences isolated subagents, and pure-Python checks gate every output.
- **Guardrails**: Outputs must pass JSON Schema, verbatim-quote and node-reference gates; failures go back to the same agent with the errors for self-correction. Per-agent tool allowlists, PreToolUse path guards, 12-turn and $3 caps.
- **Reliability**: Resumable runs via a state machine; re-analysis snapshots prior results so a failed run never leaves a story worse. Transient 5xx errors back off; quota limits fail fast.
- **Memory**: Discussion transcripts are distilled into gated conclusions and recalled in later sessions; idempotent background settling means a session is never billed twice.
- **Cost & latency**: Per-turn token ledger with a 70% prompt-cache hit rate, about $0.70–0.90 per critique; disabling extended thinking cut reply time by 75% (83.6 s to 20.3 s).
- **Full stack**: FastAPI streams progress over SSE to a React UI that draws the analysis as an intent graph and tension curve. 478 Python and 156 front-end tests, 100% coverage gate on the deterministic layer.

Software Engineer Oct 2021 – Apr 2024
Openfind *Taipei, Taiwan*

- **Legacy system modernization**: Refactored 30+ legacy C CGI programs into shared libraries and built a REST API layer on top, so the web UI, mobile app and customer integrations share one backend.
- **Exchange migration**: Owned a C# Windows client that syncs Outlook mail, contacts and calendars with Mail2000 over MAPI (Redemption library), used by 10,000+ end users at customers migrating off Exchange.
- **Two-factor authentication (2FA)**: Owned the login integration with a partner's authentication API, supporting OTP codes and push-notification approval.
- **High availability**: Owned the failover module, Perl scripts driving keepalived (VRRP), lsyncd file replication and MongoDB replica sets; the standby takes over within ~10 seconds.

Projects

randyxu0711.github.io | *three.js / WebGL, Jekyll, GitHub Actions, Python* 2026

- Three scenes in three.js, incl. a 270-book shelf in one instanced draw call; plain HTML fallback.
- Serverless CMS on GitHub Actions: opening an issue fetches the page metadata, archives it on the Wayback Machine and deploys; a weekly job points dead links to the archive. 216 tests.

wpSBOOT | *Python, FastAPI, PostgreSQL, Docker* 2019, rebuilt 2026

- Web server for a phylogenetic method published in *Bioinformatics* (NCCU lab), re-architected in one day with an AI coding agent; PostgreSQL job queue with `SKIP LOCKED` and worker heartbeats. 59 tests.

Education

National Chengchi University | B.S. in Computer Science 2016 – 2020